Tianruo Rose Xu

Mobile: +1 607-262-1199 Email: tx88@cornell.edu Website: xtrose29.github.io

Education

Cornell University, College of Engineering Aug 2024 – May 2027
B.S. in Computer Science Minor in Mathematics

Honors and Awards

- Cornell Computer Science nomination, CRA Outstanding Undergraduate Researcher Award, 2027.
- Hunter R. Rawlings III Cornell Presidential Research Scholar.
- Clare Boothe Luce Undergraduate Research Award.
- Menschel Cornell Commitment Public Service Grant.
- Cornell Graduate School Opportunity Fund Grant.
- Bowers Undergraduate Research Experience (BURE) stipend and Academic Year Expansion support.
- USA Computing Olympiad (USACO) Platinum Division; full score in the January 2023 Gold Contest.
- USA Mathematical Olympiad (USAMO) Qualifier.

Publications and Manuscripts

[**Preprint**] **Tianruo Rose Xu**, Jiawei Ren, Yichi Yang, Zhaoxu Zheng, Lianhui Qin. *RT-SAFE: Benchmarking Agent Safety in Real-Time Embodied Environment.* arXiv preprint arXiv:2610.09294, October 2026. [Paper] [Project Website] [Code].

[**Preprint**] Haoqiang Kang, Yiming Zhang, Yiyang Guo, Chuying Li, Jianzhi Shen, **Tianruo Rose Xu**, Xiaokang Ye, Lianhui Qin. *DeliveryGym: An RL Environment for Long-Horizon Embodied Agent Planning with Adaptive Curriculum.* arXiv preprint arXiv:2609.19801, September 2026. [Paper] [Code].

[**Submitted**] *Can Emergent Verbalization of Epistemic Uncertainty Inform Factuality?*

[**Published**] **Tianruo Rose Xu**. *Can You Trust What I Think? Analyzing and Improving Verbalized Uncertainty and Factuality in Reasoning-Based Large Language Models.* AAAI-26 Undergraduate Consortium, Singapore, January 2026. [Poster] [Oral Presentation] [Paper].

[**Published**] **Tianruo Rose Xu**, Vedant Gaur, Liu Leqi, Tanya Goyal. *The Progress Illusion: Revisiting Meta-Evaluation Standards of LLM Evaluators.* Findings of the Association for Computational Linguistics: EMNLP 2025, Suzhou, China, November 2025. [Paper] [Poster].

[**Published**] Yingke Ding, Jiankai Tang, Wanying Mo, **Tianruo Rose Xu**, Yuanchun Shi, Yuntao Wang. *NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model.* MICCAI 2025 Workshop MSB EMERGE, July 2025. [Paper] [Code]. Runner-up Best Paper.

[**Published**] **Tianruo Xu**. *Elder Depression Detection by Multimodal Means.* BCP Education & Psychology, November 2022. [Paper].

Research Experience

Undergraduate Researcher, Cornell University

Computer Science Department, NLP Group; advised by Professor Tanya Goyal and Professor Claire Cardie

Verbalized Epistemic Uncertainty and Factuality in Reasoning-Based LLMs 2025 – Present

- Investigate whether verbalized epistemic uncertainty in LLM reasoning traces predicts claim-level factuality in long-form responses.
- Led project design with Anirudh Atmakuru; built a pipeline for reasoning-to-claim alignment, FActScore evaluation, and factuality classifier training.
- Identified signals of factual reliability in reasoning traces and evaluated their utility for detecting potentially incorrect claims.

The Progress Illusion: Meta-Evaluation of LLM Evaluators 2025

- Investigated how meta-evaluation choices affect conclusions about the quality and progress of LLM-based evaluators.
- Led experiments measuring agreement with human judgments and sensitivity to reference-model selection and evaluation settings.
- Demonstrated that apparent evaluator improvements can depend strongly on the meta-evaluation setup; contributed analysis, visualizations, and manuscript writing.

Content and Style Evaluation in Natural Language Generation **2024**

- Designed scoring pipelines, error analyses, and ablations to study content quality and writing style as distinct dimensions of generation evaluation.

Research Intern, UC San Diego

Computer Science Department, Q-Lab; advised by Professor Lianhui Qin

RT-SAFE: Real-Time Embodied Agent Safety **2026 – Present**

- Led RT-SAFE, a benchmark for embodied-agent safety in environments that continue evolving during model inference; designed tasks, safety metrics, and the real-time evaluation protocol.
- Led frontier vision-language model experiments; found **12.3**× more collisions under real-time than static evaluation, with **94.1%** goal completion but only **0.7%** safety-event-free episodes.
- Analyzed how reasoning effort and inference latency affect safety, showing that longer inference can increase exposure to dynamic hazards.
- Led preprint analysis and writing, released evaluation code, and extended comparisons to newly released model versions.

DeliveryGym / DeliveryBench: Embodied Agent Environments **2026**

- Developed dynamic traffic, obstacle, and safety infrastructure and maintained the Unreal Engine first-person observation pipeline for embodied-agent evaluation.
- Integrated the CityCore Paris environment with semantic points of interest, navigation graphs, and map assets to support urban navigation and safety tasks.

Undergraduate Researcher, Cornell University

Mathematics Department, Math + AI Lab; advised by Professor Daniel Halpern-Leistner

MathCopilot / AWMBench: Evolving Benchmarks for Formal Mathematics 2026 – Present

- Develop solver and evaluation infrastructure for research-level mathematical proof completion within the MathCopilot LaTeX-to-Lean pipeline.
- Implement whole-file prompting and Lean-feedback proof-repair baselines behind a shared solver interface, with independent checks of proof validity and theorem-statement preservation.
- Build a reproducible benchmark runner and leaderboard with versioned inputs, fresh-cache trials, and tracking of proof completion, runtime, token usage, and cost.

Independent Research Collaboration, Oracle **2025 – Present**

Remote; mentored by Priyan Pattnayak, Senior Principal Scientist at Oracle Cloud.

- Conduct research on uncertainty and factuality in reasoning-based language models, with mentorship supporting the AAAI-26 Undergraduate Consortium work.
- Design controlled ASL video-question-answering experiments with correct, corrupted, and missing translations to study textual influence on visually grounded answers.
- Develop an evaluation pipeline separating error detection, evidence selection, and answer repair; compare frame density, targeted reinspection, and retention of prior textual context.

Undergraduate Research Assistant, Tsinghua University

Department of Computer Science and Technology, Pervasive Human-Computer Interaction Lab.

Intelligent AI Tool Benchmarking and Orchestration Platform **2025 – 2026**

- Investigate capability-based tool representations and orchestration for matching external tools to changing user goals.
- Designed and implemented back-end retrieval and matching for user intents, tool capabilities, and personalized tool selection.

NFCMTL: Auto NailFold Capillaroscopy through a Multi-Task Learning Model **2025**

- Preprocessed data, trained baselines, and evaluated segmentation, classification, and keypoint detection for a multi-task nailfold capillaroscopy model.

Undergraduate Research Assistant, Cornell University **2024**

Information Science Department; advised by Keigo Kusumegi and Professor Yian Yin.

- Processed large-scale Wikipedia data and applied K-means clustering to analyze relationships between editor collaboration patterns and article quality.

Teaching Experience

Cornell University

- **Head Teaching Assistant**, CS 3780: Introduction to Machine Learning — Fall 2026
  Designed the Pokéverse Kaggle challenge, in which students predict battle advantage scores from pairs of noisy images using partially labeled character identities; developed the dataset, evaluation metric, and model baselines.
- **Climbing Instructor**, PE 1640: Basic Rock Climbing — Fall 2026
  Certified in First Aid and CPR.
- **Teaching Assistant**, CS 3780: Introduction to Machine Learning — Spring 2026
  Designed the final Kaggle challenge, All About Birds!, a bird-sound spectrogram classification task with course-aligned baselines and evaluation.
- **Teaching Assistant**, CS 4820: Introduction to Analysis of Algorithms — Fall 2025
- **Teaching Assistant**, CS 2800: Mathematical Foundations of Computing — Spring 2025